Project 6:- Maven Selenium Project – MyMavenSeleniumApp04

Objective

This project involves writing a Selenium script that operates across three different websites in separate browser tabs, automating login forms, a product search, and user authentication all within a single test run.

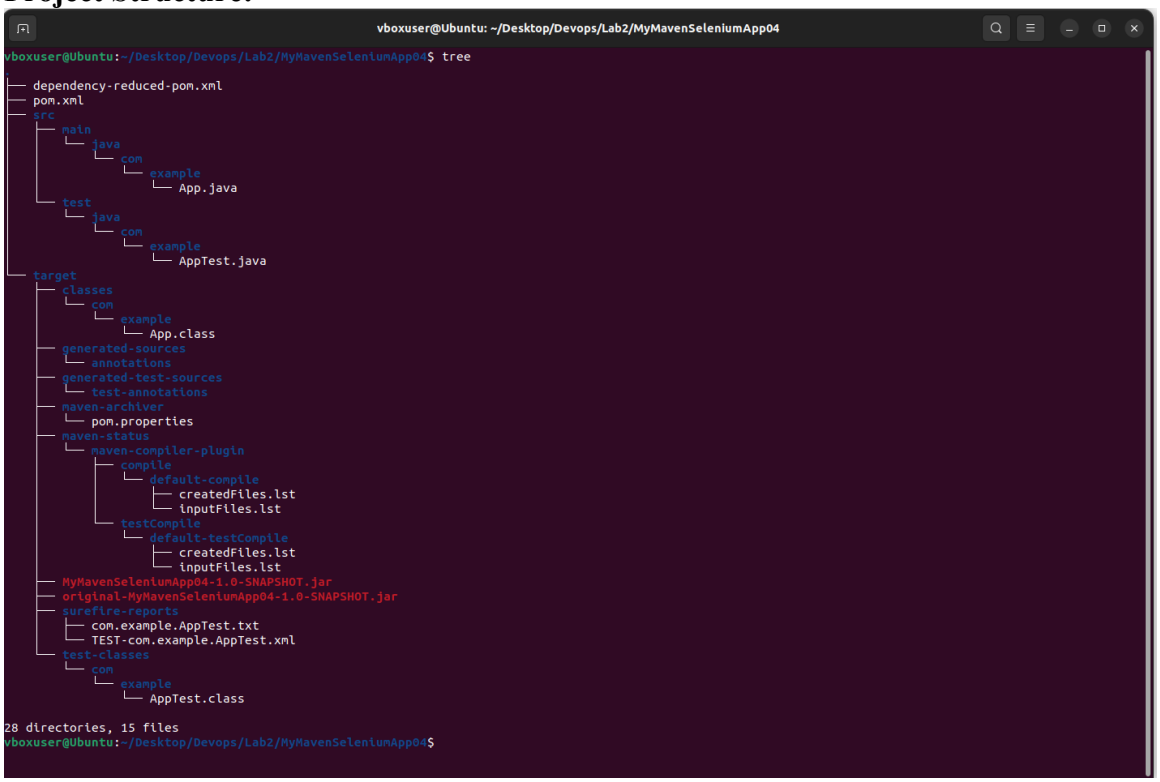
Websites Used for Automation

1. https://www.saucedemo.com/
2. https://automationexercise.com/products
3. https://practicetestautomation.com/practice-test-login/

Step 1: Create a Maven Project Using Terminal

```
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp04
  -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

Project Structure:



Step 2: Add Selenium Dependency in pom.xml

```
<dependency>
  <groupId>org.seleniumhq.selenium</groupId>
  <artifactId>selenium-java</artifactId>
  <version>4.41.0</version>
</dependency>
```

Step 3: Inspect Elements in the Websites

Inspect each site using DevTools and note the following HTML element IDs that the script will reference:
SauceDemo: user-name, password, login-button
Automation Exercise: search_product, submit_search

Practice Test Automation: username, password, submit

Step 4: Create App.java

Open App.java at src/main/java/com/example/ and replace the contents with the code below:

```
package com.example;


import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.WindowType;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;

public class App {
  public static void main(String[] args) throws InterruptedException {
    ChromeOptions options = new ChromeOptions();
    options.addArguments("--start-maximized");
    WebDriver driver = new ChromeDriver(options);
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));

    driver.get("https://www.saucedemo.com/");
    driver.findElement(By.id("user-name")).sendKeys("standard_user");
    driver.findElement(By.id("password")).sendKeys("secret_sauce");
    driver.findElement(By.id("login-button")).click();

    driver.switchTo().newWindow(WindowType.TAB);
    driver.get("https://automationexercise.com/products");
    driver.findElement(By.id("search_product")).sendKeys("Men Tshirt");
    driver.findElement(By.id("submit_search")).click();
    WebElement product = wait.until(
      ExpectedConditions.visibilityOfElementLocated(
        By.cssSelector("a[data-product-id='2']")));
    product.click();

    driver.switchTo().newWindow(WindowType.TAB);
    driver.get("https://practicetestautomation.com/practice-test-login/");
    driver.findElement(By.id("username")).sendKeys("student");
    driver.findElement(By.id("password")).sendKeys("Password123");
    driver.findElement(By.id("submit")).click();
```

```
        Thread.sleep(5000);
        driver.quit();
    }
}
```

Step 5: Build the Maven Project

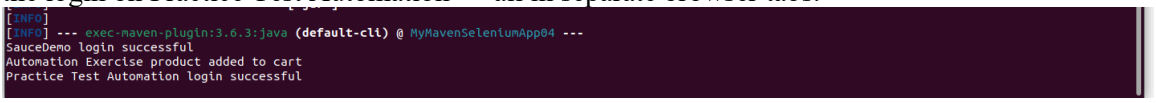
```
mvn clean compile
```

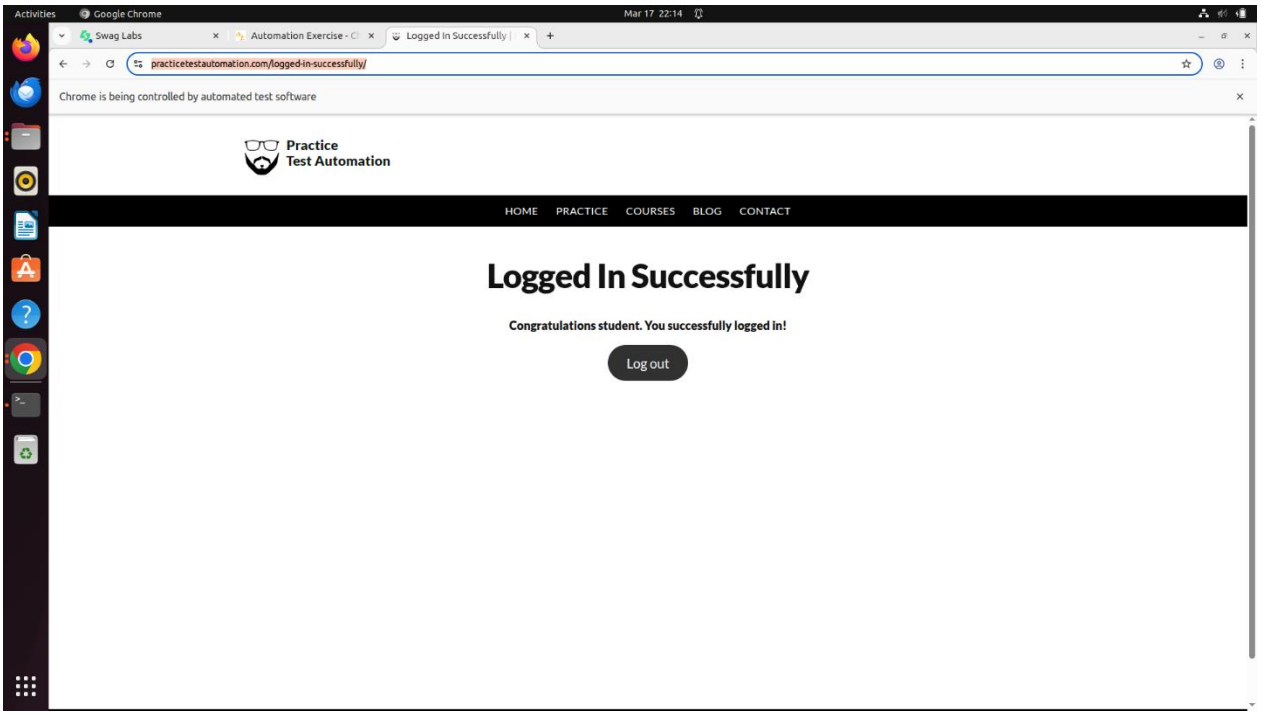
Step 6: Run the Application

```
mvn exec:java
```

Output

The script successfully logs in to SauceDemo, performs a product search on Automation Exercise, and completes the login on Practice Test Automation — all in separate browser tabs.



Conclusion

Working across multiple sites in a single session proved that Selenium WebDriver is well-suited for complex, multi-step automation. Tab management and explicit waits made the script stable and reliable.